Ifeanyi Desmond Dike

Senior Software Engineer | Voice AI & Distributed Systems

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PROFILE

Senior software engineer specializing in **voice AI and distributed backend systems**. Builds production call infrastructure and asynchronous processing pipelines with Python, Go and TypeScript. Experience includes telephony, streaming speech, LLM orchestration, capacity management and multi-caller reporting, alongside applied research into full-duplex voice-agent evaluation.

RESEARCH & PUBLICATIONS

ATR: Acoustic Turn-over Rate | Sole-author preprint, 2026
Developed an ASR-independent metric and evaluation pipeline for overlap handling in full-duplex voice agents. Released code.

Full-Duplex-Bench studies | Two sole-author preprints, 2026
Reproduced Gemini voice-agent evaluations on Apple Silicon: ASR backend sensitivity and overlap handling and prosodic adaptation. Released the evaluation pipeline.

IEEE IEMCON 2018: Co-authored four papers on antenna modeling, biometric attendance, electronic locking and university clearance systems. Publication list.

PROFESSIONAL EXPERIENCE

DataStrut AI

Founding AI Engineer / Senior Software Engineer | Aug 2025 - Present | Remote

- Own the production voice platform across telephony, streaming speech, LLM orchestration and post-call processing; migrated managed voice infrastructure to a self-hosted Python/Pipecat stack.
- Implemented call admission and an on-demand Fly Machines pool with atomic Redis reservations, reserved inbound capacity and safe drain/resume controls to protect active calls.
- Built role-aware, multi-caller reporting with PostgreSQL data-model changes to consolidate input from multiple callers into a shared report.
- Separated outbound scheduling from report generation with dedicated RQ workers. Added asynchronous, idempotent call triggers and capacity-aware retries to avoid duplicate dials and queue contention.
- Load-tested event ingestion with **6,000 successful synthetic requests** across four load levels. Modeled CPU, memory and report capacity; identified Redis connections as the first call-routing constraint.
- Cut steady-state STT sessions by **about 67% (3 to 1 per call)** in three-agent workflows through activation-based lifecycles. Delivered English/Spanish recognition, synthesis, prompts and reporting.
- Built response-admission controls, echo-aware turn-taking and deterministic regression tests; added playback-aligned traces and recordings to distinguish generated speech from what callers heard.

Cenphi

Co-Founder and Lead Developer | Mar 2025 - Present | Remote

- Built an asynchronous video-production pipeline with Python, FastAPI, Whisper, diarization, LLMs and FFmpeg; added job orchestration, provider fallback and live progress updates.
- Developed segment-based story composition and artifact tracking, reducing manual review by **approximately 80%**.

PROFESSIONAL EXPERIENCE - CONTINUED

Fynance Hub

Technical Lead | Oct 2024 - Present | Remote, part-time

- Built Fynance Hub's mortgage lead-generation product, which has generated **hundreds of leads for Canadian mortgage agents** and streamlined how they find and qualify prospects.
- Developed GPT-4/RAG recommendations for broker leads and automated outbound voice verification.
- Designed Go/Python microservices with gRPC, PostgreSQL and Redis for real-time data processing.
- **Separate project:** building MajorDairy AI's app for large Canadian farms, currently in **Google Play testing**. The company has raised **CAD 500K in seed investment**.
- Developing React Native/Expo and FastAPI/PostgreSQL workflows for breeding, pregnancy checks, calving and daily farm, technician and vet tasks.
- Built a DairyComp 305 importer to bring legacy herd records into the mobile platform.

Arbor Platform Inc.

Senior Software Engineer | Feb 2023 - Oct 2024 | Remote

- Built Claude/OpenAI-powered video workflows that reduced production time by **60%** and improved throughput by **80%**.
- Delivered text-to-video and audio-transcription features for AI-assisted editing.
- Built a content studio and B-roll asset-management tools with one-click social distribution.
- Developed React/TypeScript interfaces and integrated backend APIs and cloud services.

Howdy Learning

Senior Software Development Engineer | May 2020 - Mar 2023 | Remote

- Reduced WebSocket network traffic by **more than 99%** through real-time communication optimizations.
- Built live-video infrastructure supporting up to **10,000 concurrent connections**.
- Developed a GetStream/WebSocket chat system serving approximately **100 communities**.
- Delivered embedded video, voice and SSO widgets for Thinkific and WordPress, with integrations used by **more than 100 customers**.

Bujuwa Nigeria Limited

Software Developer | Jul 2018 - Dec 2019 | Abuja, Nigeria

- Delivered web and mobile assessment systems, from requirements and architecture to implementation.

TECHNICAL SKILLS & OPEN SOURCE

Backend: Python, asyncio, Go, TypeScript, FastAPI, gRPC, PostgreSQL, Redis, MongoDB, RQ.

Voice & AI: Pipecat, Twilio, WebSockets, WebRTC, streaming STT/TTS, VAD, turn-taking, LLM tools, RAG, evaluation, PyTorch, Hugging Face.

Infrastructure: Fly.io, Docker, Kubernetes, AWS, GCP, Azure, CI/CD, observability, distributed systems.

Pipecat: Reported and documented an Anthropic integration bug (#5035 and #5036); linked to the maintainer-authored fix in PR #5041.

EDUCATION

M.Sc., Financial Engineering | WorldQuant University | 2026 - Present (in progress)

B.Tech., Applied Mathematics, First Class | Federal University of Technology, Akure | 2018
GPA: 4.63/5.00. National university mathematics competition: Gold (2017), Bronze (2016).